

C.W

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Function

• Suppose,

$$y = x^2$$

- x is value independent variable
- y is dependent on x

for base based x is

Def: [from books or online]

• Suppose,

$$y = |x|$$

x	$+3$	-3	
y	$+3$	$+3$	

$$x \in \mathbb{N} \quad (1 \rightarrow \infty)$$

$$x \in \mathbb{R} \quad (0 \rightarrow \infty)$$

$$x \in \text{int} \quad (-\infty \rightarrow +\infty)$$

Again,

$$x = \sqrt{y}$$

y	4	9	
x	$+2$	-2	

• अतः,

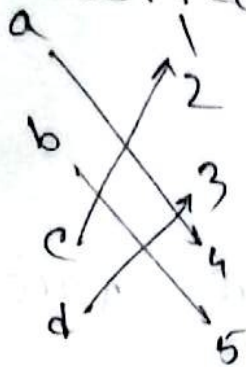
- $\sqrt{\cdot}$ Domain $0 \rightarrow \infty$

same range $0 \rightarrow \infty$

Domain $\sqrt{\cdot}$ is $\mathbb{R}^+$

co-domain.

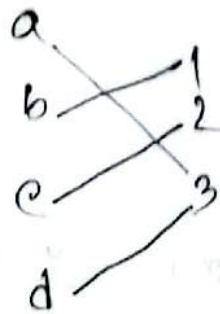
Practice: (find one-to-one or onto)



Case-1

one-to-one ☒

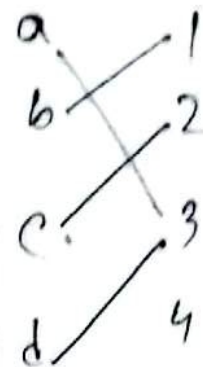
onto ☐



Case-2

one-to-one ☒

onto ☒



one-to-one ☐

onto ☒

(*) Composition of function: (for 4 marks)

Suppose;

$$f(x) = 2x + 3$$

$$g(x) = 3x + 2$$

$$\therefore f(g(x)) = 2(3x + 2) + 3$$

$$= 6x + 4 + 3$$

$$= 6x + 7$$

$\Rightarrow f \circ g$

$$\therefore g \circ f = g(f(x))$$

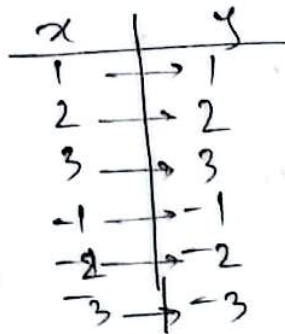
$$= 3(2x + 3) + 2$$

$$= 6x + 9 + 2$$

$$= 6x + 11$$

(*) One to one:

$$y = x$$



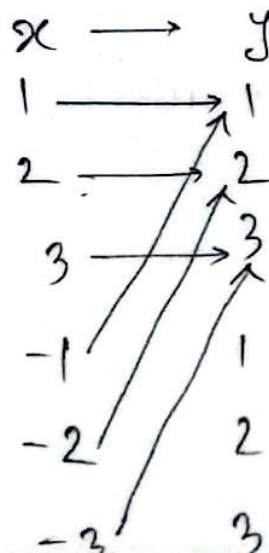
• ଯାଦି y ଯେକୌଣସି x

ହେଉଥିଲେ One to one.

• ଯଦି x ଯେକୌଣସି y ନୁହେଁ ତେବେ not one to one.

(*) Onto:

Suppose,
 $y = |x|$



• ଯାଦି y ଯେକୌଣସି x

ହେଉଥିଲେ onto.

• ଯଦି x ଯେକୌଣସି y ନୁହେଁ ତେବେ not onto.